The Number-Density Hinge

The same counting structure, three different physical consequences

Name: ____________________

Class: ____________________

IB DP Physics: C.3, B.5 and B.3

Working rule: A density states how many objects occupy a unit length or a unit volume. It does not by itself determine a diffraction angle, an electric current, or a gas pressure. A system-specific physical relation supplies the consequence.

1. Diffraction grating: line density becomes spacing

A diffraction grating has 600 lines per millimetre. Monochromatic light of wavelength $\lambda = 500\,\mathrm{nm}$ is incident normally. A screen is $D = 1.50\,\mathrm{m}$ from the grating.

A. Turn a count into a spacing

Sketch a short section of the grating and mark the separation d between neighbouring lines. Convert the grating density into lines per metre and calculate

$$d = \frac{1}{N_g}.$$

State the units of N_g and d.

B. Supply the interference law

For principal maxima,

$$d \sin\theta = m\lambda.$$

Calculate the angles of the first- and second-order maxima. Explain why the line density alone cannot determine either angle.

C. Locate the maxima

Use $y = D\tan\theta$ to calculate the distance of each first-order maximum from the central maximum. State why there are two first-order maxima. On a labelled sketch, show $m = 0$, $m = +1$ and $m = -1$.

D. Change the density

The number of lines per millimetre is doubled while λ is unchanged. Predict the changes to d and the first-order angle. Which relation contains the geometrical count, and which relation turns it into an optical prediction?

2. Electric current: carrier density becomes charge flow

A wire has mobile-electron number density $n_c = 8.5\times10^{28}\,\mathrm{m^{-3}}$, cross-sectional area $A = 1.0\,\mathrm{mm^2}$ and electron drift-speed magnitude $v_d = 1.0\times10^{-4}\,\mathrm{m\,s^{-1}}$. Take $e = 1.60\times10^{-19}\,\mathrm{C}$.

A. Count the carriers crossing a section

In time Δt, the drifting electrons sweep through a cylinder of length $v_d\Delta t$ and volume $Av_d\Delta t$. Use the number density to write an expression for the number of electrons crossing the section. Add the charge transferred.

B. Build and apply the current relation

Starting from $I = \Delta Q/\Delta t$, show that the current magnitude is

$$I = n_c e A v_d.$$

Calculate the current. Include the conversion from $\mathrm{mm^2}$ to $\mathrm{m^2}$.

C. Interrogate the factors

Predict the effect on I of separately doubling: (i) n_c, (ii) A, and (iii) v_d. Explain why the number density does not determine the current by itself. Which factor contains information about the carrier species?

D. Direction and meaning

Mark electron drift and conventional current on a wire diagram. Why are their directions opposite? Distinguish the number of carriers available per unit volume from the number crossing a section per unit time.

3. Ideal gas: particle density becomes pressure

A rigid container of volume $V = 1.00\times10^{-2}\,\mathrm{m^3}$ contains $N = 2.40\times10^{23}$ ideal-gas particles at $T = 300\,\mathrm{K}$. Take $k_\mathrm{B} = 1.38\times10^{-23}\,\mathrm{J\,K^{-1}}$.

A. Define the particle number density

Calculate

$$n_\mathrm{gas} = \frac{N}{V}.$$

State its unit. Explain the difference between N, the total number of particles, and n_gas, the number per unit volume.

B. Supply the equation of state

Use

$$PV = Nk_\mathrm{B}T \qquad \text{or} \qquad P = n_\mathrm{gas}k_\mathrm{B}T$$

to calculate the pressure. Identify the quantities that the number density alone does not supply.

C. Read the proportionality

Rearrange the ideal-gas equation to obtain

$$\frac{N}{V} = \frac{P}{k_\mathrm{B}T}.$$

Predict the change in number density when: (i) P doubles at constant T; (ii) T doubles at constant P; (iii) the same N is compressed to half the volume.

D. Interrogate the model

Explain why $N/V \propto P/T$ is a statement about an ideal gas rather than a definition of number density. What model assumptions allow the state of the gas to be represented by P, V, N and T?

Synthesis: density organises a count; the physical law supplies its effect

1. Complete the comparison.

System	Density	Unit	Further relation	Observable consequence
Grating				
Wire				
Ideal gas				

2. Develop the central claim.
Respond to:

"A density tells us how many are packed into space. It does not tell us what that population will do."

Use all three cases. Include the terms *line density*, *spacing*, *number density*, *charge carrier*, *drift speed*, *equation of state*, *physical law*, and *observable*. Explain why quantities with the same "count per space" structure can lead to very different physics.

Teacher guide and indicative answers

The Number-Density Hinge

1. Grating: a geometrical density

A. Spacing.

$$N_g = 600\,\text{mm}^{-1} = 6.00 \times 10^5\,\text{m}^{-1},$$

$$d = \frac{1}{N_g} = 1.67 \times 10^{-6}\,\text{m}.$$

N_g has unit m^{-1}; d has unit m.

B. Angles.

$$\sin\theta_1 = \frac{500 \times 10^{-9}}{1.67 \times 10^{-6}} = 0.300, \quad \theta_1 = 17.5^\circ.$$

$$\sin\theta_2 = 0.600, \quad \theta_2 = 36.9^\circ.$$

The line density only gives d. The wavelength and order m are also required.

C. Screen position.

$$y_1 = (1.50)\tan 17.5^\circ = 0.473\,\text{m}.$$

There are symmetric $m = +1$ and $m = -1$ maxima on opposite sides of $m = 0$.

D. Doubling density. $N_g \to 2N_g$ gives $d \to d/2$. For fixed λ and $m = 1$, $\sin\theta$ doubles, so the first-order maximum moves farther from the centre provided the order remains possible. $d = 1/N_g$ converts count to spacing; $d\sin\theta = m\lambda$ supplies the optical prediction.

Syllabus boundary. No small-angle approximation, resolving-power formula, or multi-slit intensity derivation is required.

2. Current: a transported population

A. Swept volume.

$$\Delta V = Av_d\Delta t, \qquad \Delta N = n_c Av_d\Delta t.$$

The magnitude of charge crossing is

$$\Delta Q = e\Delta N = n_c eAv_d\Delta t.$$

B. Current.

$$I = \frac{\Delta Q}{\Delta t} = n_c eAv_d.$$

Since $1.0\,\text{mm}^2 = 1.0 \times 10^{-6}\,\text{m}^2$,

$$I = (8.5 \times 10^{28})(1.60 \times 10^{-19})(1.0 \times 10^{-6})(1.0 \times 10^{-4}) = 1.36\,\text{A}.$$

C. Factors. Doubling any one of n_c, A or v_d doubles I when the others are fixed. Density gives carrier availability; current also requires cross-section, drift speed and charge per carrier. The factor q or e identifies the charge carried by each particle.

D. Direction. Electrons drift opposite to conventional current because their charge is negative. Number density is a spatial count; current is a rate of charge crossing a surface.

Syllabus boundary. The task uses the IB relation $I = nqAv_d$ without introducing mobility, conductivity, relaxation time or microscopic band theory.

3. Gas: density in an equation of state

A. Number density.

$$n_{\text{gas}} = \frac{2.40 \times 10^{23}}{1.00 \times 10^{-2}} = 2.40 \times 10^{25}\,\text{m}^{-3}.$$

N counts the particles in the whole container; n_{gas} gives the count per unit volume.

B. Pressure.

$$P = n_{\text{gas}}k_{\text{B}}T = (2.40 \times 10^{25})(1.38 \times 10^{-23})(300) = 9.94 \times 10^4\,\text{Pa}.$$

Number density alone does not supply pressure: temperature and the ideal-gas model are also needed.

C. Proportionality.

$$n_{\text{gas}} = \frac{P}{k_{\text{B}}T}.$$

At fixed T, doubling P doubles n_{gas}. At fixed P, doubling T halves n_{gas}. Halving V at fixed N doubles n_{gas}.

D. Model. The first equality $n_{\text{gas}} = N/V$ is a definition. The relation $n_{\text{gas}} = P/(k_{\text{B}}T)$ follows only from the ideal-gas equation of state. Expected assumptions include a large collection of particles, negligible particle volume relative to the container, and negligible intermolecular forces except during collisions.

Syllabus boundary. Do not infer mean particle spacing, mean free path, or a Maxwell–Boltzmann speed distribution.

Teacher synthesis and assessment focus

System	Density	Unit	Further relation	Consequence
Grating	$N_g = \text{lines}/L$	m^{-1}	$d = 1/N_g$, $d\sin\theta = m\lambda$	maxima angles and positions
Wire	$n_c = N/V$	m^{-3}	$I = n_c qAv_d$	charge flow per unit time
Ideal gas	$n_{\text{gas}} = N/V$	m^{-3}	$P = n_{\text{gas}}k_{\text{B}}T$	pressure for a stated temperature

Indicative central response. A density is a quotient that records population per spatial extent. In the grating it is an engineered line count whose inverse is the slit spacing; interference then determines the maxima. In a wire, carrier density enters a transport relation with carrier charge, cross-sectional area and drift speed. In an ideal gas, particle density enters an equation of state with temperature. The same mathematical structure therefore does not identify the mechanism or observable.

Assessment emphasis: correct units; clear distinction between total number and number density; identification of the additional physical relation; proportional reasoning; explanations that do not treat density itself as a causal mechanism.